

Tic Tac Toe Game in C++

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#include <iostream>
using namespace std;

char board[3][3] = {
    {'1', '2', '3'},
    {'4', '5', '6'},
    {'7', '8', '9'}
};

char currentPlayer = 'X';

void displayBoard() {
    cout << "\n";

    for (int i = 0; i < 3; i++) {
        cout << " ";

        for (int j = 0; j < 3; j++) {
            cout << board[i][j];

            if (j < 2)
                cout << " | ";
        }

        cout << "\n";

        if (i < 2)
            cout << "---|---|---\n";
    }

    cout << "\n";
}

bool placeMark(int choice) {
    int row = (choice - 1) / 3;
    int col = (choice - 1) % 3;

    if (board[row][col] != 'X' && board[row][col] != 'O') {
        board[row][col] = currentPlayer;
        return true;
    }

    return false;
}

bool checkWin() {
    // Rows
    for (int i = 0; i < 3; i++) {
        if (board[i][0] == currentPlayer &&
            board[i][1] == currentPlayer &&
            board[i][2] == currentPlayer)
            return true;
    }

    // Columns
    for (int i = 0; i < 3; i++) {
        if (board[0][i] == currentPlayer &&
            board[1][i] == currentPlayer &&
            board[2][i] == currentPlayer)
            return true;
    }

    // Diagonals
    if (board[0][0] == currentPlayer &&
        board[1][1] == currentPlayer &&
        board[2][2] == currentPlayer)
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        return true;

    if (board[0][2] == currentPlayer &&
        board[1][1] == currentPlayer &&
        board[2][0] == currentPlayer)
        return true;

    return false;
}

bool checkDraw() {
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            if (board[i][j] != 'X' && board[i][j] != 'O')
                return false;
        }
    }

    return true;
}

void switchPlayer() {
    if (currentPlayer == 'X')
        currentPlayer = 'O';
    else
        currentPlayer = 'X';
}

int main() {
    int choice;

    cout << "==== TIC TAC TOE =====\n";

    while (true) {
        displayBoard();

        cout << "Player " << currentPlayer
              << ", enter position (1-9): ";
        cin >> choice;

        if (choice < 1 || choice > 9) {
            cout << "Invalid choice! Try again.\n";
            continue;
        }

        if (!placeMark(choice)) {
            cout << "Position already taken! Try again.\n";
            continue;
        }

        if (checkWin()) {
            displayBoard();
            cout << "Player " << currentPlayer << " wins!\n";
            break;
        }

        if (checkDraw()) {
            displayBoard();
            cout << "Game Draw!\n";
            break;
        }

        switchPlayer();
    }

    return 0;
}

```